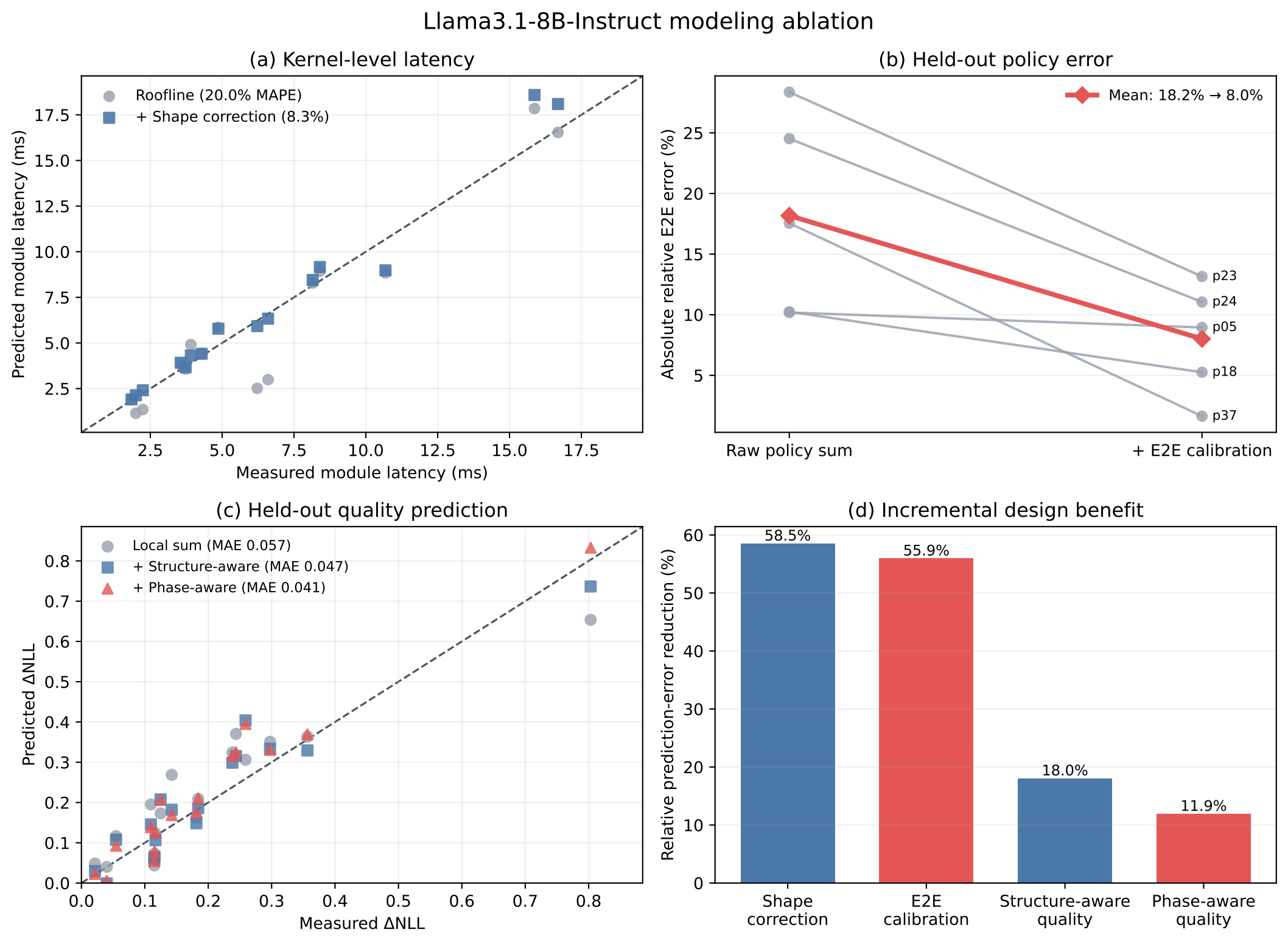
Llama3.1-8B-Instruct modeling ablation
(a) Kernel-level latency
Roofline (20.0% MAPE)
+ Shape correction (8.3%)
Predicted module latency (ms)
Measured module latency (ms)
(b) Held-out policy error
Mean: 18.2% → 8.0%
Absolute relative E2E error (%)
p23
p24
p05
p18
p37
Raw policy sum
+ E2E calibration
(c) Held-out quality prediction
Local sum (MAE 0.057)
+ Structure-aware (MAE 0.047)
+ Phase-aware (MAE 0.041)
Predicted ΔNLL
Measured ΔNLL
(d) Incremental design benefit
Relative prediction-error reduction (%)
58.5%
55.9%
18.0%
11.9%
Shape correction
E2E calibration
Structure-aware quality
Phase-aware quality